FLOODCAST

Predict the Surge Before It Strikes

THE CHALLENGE IN ONE LINE

Build a machine-learning model that predicts the amount of water flowing through a river each day, using weather information and the river area's physical features.

1. What is this challenge about?

Floods can kill people and force families to leave their homes. If we can estimate how much water will flow through a river, communities may get more time to prepare.

You are given historical river-flow measurements, rainfall and temperature information, plus details about the land around each river. Use these to train a regression model that predicts river discharge for dates in the test data.

What does “discharge” mean?

Discharge is the amount of water flowing past a point in a river every second. It is measured in **cubic feet per second (cfs)**. Your model must predict a number, not just say “safe” or “flood”. Higher discharge can mean greater flood danger.

2. What exactly must your model predict?

For every **station_id + date** combination in **test.csv**, predict one value:

discharge_cfs — the river's daily discharge in cubic feet per second. The prediction must be a non-negative number.

Example: If test.csv contains station 01646500 on 2023-08-10, your model must predict the discharge for that station on that date.

3. What data can you use?

The challenge uses three free public data sources. The organizers will also provide a pre-processed starter dataset with five stations already joined together.

DATA SOURCE	IN SIMPLE WORDS	WHAT IT ADDS
USGS river data	Past measurements from river monitoring stations.	The river discharge you are trying to predict (the target).
Open-Meteo weather	Historical weather for the location of each station.	Rainfall, maximum/minimum temperature and soil moisture.
USGS StreamStats	Information about the land area that drains into a river station.	Drainage area, elevation, slope and forest cover.

How do the datasets connect?

- Match river measurements with catchment details using **station_id**.
- Match weather to the river station using its **latitude, longitude and date**.
- Combine them into one dataset containing weather + land features + historical discharge.

The challenge brief says a pre-joined starter CSV with five stations will be provided. Teams may add more stations from the same sources.

4. What files will you receive?

FILE	WHAT IT CONTAINS
train.csv	Past examples: station, date, weather, land features and the actual discharge. Use this to train your model.
test.csv	Station, date, weather and land features — but no discharge. This is what your model must predict.
sample_submission.csv	An example output file with discharge values filled with zeros. Replace them with your predictions.
station_metadata.csv	Station name, location, state and river name.

5. What should you submit?

Create a file named **predictions.csv** with exactly three columns, in this exact order:

station_id, date, discharge_cfs

There must be exactly one prediction for every row in test.csv. Every discharge value must be a non-negative number.

Your file will be rejected if:

- A station/date pair from test.csv is missing.
- A prediction is negative or blank/NaN.
- Your file has more or fewer rows than test.csv.
- The column names or their order are wrong.
- The file is not a UTF-8 encoded CSV.

6. How will your team be judged?

Your predictions are automatically compared with the hidden actual river-discharge values. There is no manual component in the auto-score.

METRIC	WEIGHT	WHAT IT MEANS	BETTER SCORE
Overall RMSE	40%	How close predictions are across all test records.	Lower
Extreme-event RMSE	30%	How well you predict the highest-discharge days (top 10% per station).	Lower
MAE	20%	Average size of prediction errors.	Lower
R-squared (R^2)	10%	How well your model captures patterns in the data.	Higher

The big takeaway: Flood peaks matter a lot. RMSE penalizes large mistakes, and 30% of the score focuses specifically on extreme high-discharge days. A model that misses major peaks can perform badly even if its ordinary-day predictions look good.

The leaderboard initially ranks by raw RMSE. The final score is a weighted combination of the four metrics after they are normalized.

7. Important rules — avoid disqualification

Never use the actual test discharge values as model input. The test targets are hidden; trying to use them is data leakage and can disqualify the submission.

Also, do not:

• Submit negative discharge predictions.
• Leave out any test rows.
• Manually edit predictions.csv instead of generating it by running your model.
• Submit a pre-built model from before the event.
• Share predictions or code with other teams.

What is allowed?

• Any ML framework, such as scikit-learn, XGBoost, LightGBM, PyTorch or TensorFlow.
• Pre-trained models as a starting point, if declared in model_card.md.
• Adding more stations from USGS.
• Feature engineering using the provided columns.
• Teams of 1–4 members.

8. A practical way to approach the challenge

STEP 1 — Inspect the files

Understand the columns in train.csv, test.csv and station_metadata.csv.

STEP 2 — Prepare the data

Check missing values, dates, station IDs and data quality. Keep the test target out of training inputs.

STEP 3 — Explore patterns

Study how rainfall, temperature, soil moisture and catchment features relate to river discharge.

STEP 4 — Train and validate

Train regression models and evaluate them on a validation split that helps test whether the model generalizes to unseen stations.

STEP 5 — Focus on flood peaks

Check performance on high-discharge days, not only ordinary river-flow days.

STEP 6 — Predict the test set

Generate one non-negative discharge prediction for every row in test.csv.

STEP 7 — Validate the CSV

Check exact columns, order, row count, station/date coverage and non-negative values.

STEP 8 — Package and submit

Push the required files to your public GitHub repository and submit the repo link plus predictions.csv.

9. What must be submitted?

ITEM	WHAT TO INCLUDE
Public GitHub repository	All code, README and requirements.txt or environment.yml.
predictions.csv	Your model's predictions for every row in test.csv.
README.md	Team name, team members, a 3–5 sentence approach summary and instructions to run the code.

ITEM	WHAT TO INCLUDE
model_card.md	Features used, model type, train/validation RMSE and known limitations.
Presentation	3–5 slides covering Problem → Data → Model → Results → Limitations. Submit as PDF or link.

Minimum GitHub repository structure

floodcast-team-name/
■■■ README.md
■■■ requirements.txt
■■■ solution.ipynb or solution.py
■■■ predictions.csv
■■■ model_card.md

10. Event timeline

TIME	WHAT HAPPENS
Hour 0	Problem statement, datasets and starter CSV are released.
Hour 1	Starter notebook walkthrough by organizers.
Hour 6	Optional organizer check-in.
Hour 20	Submission deadline: GitHub link + predictions.csv. No late submissions.
Hour 22	3-minute team presentations begin.
Hour 24	Winners announced.

The simplest way to remember FLOODCAST

Use past river data + weather + land features → train a regression model → predict river discharge → submit accurate predictions.

This is a simplified guide to the supplied FLOODCAST Participant Problem Statement (Version 1.0, November 2026). For exact dataset fields, scoring rules, file formats and submission instructions, follow the original challenge brief.

Source links listed in the original brief: USGS National Water Information System — waterdata.usgs.gov/nwis/dv · Open-Meteo Historical Weather API — open-meteo.com/en/docs/historical-weather-api · USGS StreamStats — streamstats.usgs.gov/ss/